



Modular Athletic Exoskeleton Project

Sensor Enclosure Placement Explorations - Report I

The document below shows the testing of materials and design choices for the placement of PCBs and recommendations thus far on how they should be attached.

The testing began with investigating the different available straps and mechanisms by which to hold the PCBs in place, then moved to where they would be located, taking into account product goals and end-user use cases, with ergonomics, design life, and successful product use as main concerns.

INTRODUCTION

The first concern was the material of the bodysuit. It seems breathable and capable of stretching, and is expected to be inoffensive to an athlete. However, it was not tested under strenuous or extended activity (yet). Concerns of sweat buildup, chafing, a decrease in elasticity over time, and other unforeseen concerns have not been tested.

Once the bodysuit was deemed acceptable, the method of attachment of the PCBs became the next concern. Initially, a hydrogel was considered, but that can wear easily over time, and it was verified that the goal of the project was current motion data, not biometric data; this means that the concern of skin-to-sensor contact is a low priority. Other options include elastic pockets and two different kinds of attachment straps. Restrictions on what materials could be used and modified at this time meant that the elastic pocket concept was not available for testing. Both types of straps were evaluated in terms of what would be best for the product.

STRAP SYSTEMS

One strap is an elastic hook and loop system (commonly referred to as VELCRO) of variable length, while the other is a buckle with length adjustment capabilities. They will be called strap systems 1 and 2, respectively.

Strap system 1 comes in a variety of lengths (two are pictured in Figure 1), but is also individually length-adjustable, and can be linked together (Figure 2). The smallest available length cannot be linked with any of the longer ones, as the smallest length strap has a smaller width as well (Figure 3).

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Figure 1



Figure 2



Figure 3

There are some benefits to the strap system 1. The hook and loop design prevents having hanging lengths of unused strap, and is more elastic than system 2 (seen later). Additionally, the varied lengths and elasticity allow for a wide range of strap sizes. They are also the lightest system that has been tested. The elasticity also allows for greatly varied tightness without discomfort. For example, the shortest strap was tested on the wrist, and a wide range of tightness was achieved without any discomfort (Figures 4 and 5). In the figures, the loosest setting fits snugly around the wrist with little compression (Figure 4), while the tightest has considerable compression, with the tail end of the strap stretching an additional half-rotation around the wrist. This elasticity also notably accommodates the flexion of the potential

athlete's muscles, staying tight when certain muscles are relaxed, and not straining when they are flexed.



Figure 4

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However, there are concerns. While the straps *can* be tightened without discomfort, the skin can be uncomfortably pinched if the applicant does not make an effort (however minimal) to prevent it (Figure 5). Another concern is the uncomfortable nature of having a buckle on or near a joint. This can be avoided with correct application, but it remains a concern as design continues.

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Figure 5

The pinching issue goes down considerably when the straps are over the suit, and should the straps be sewn onto/into the suit, the designer can make sure to place them carefully.

Moving on to strap system 2, many of the benefits remain. However, they only come in one very long length and can only get so short in their initial state. This could hypothetically be fixed by simply cutting them shorter. They require much more force to stretch than the strap system 1, which could become a desirable trait further in the design process, depending on the results of further investigation into the desires of the end user. Most notably, the buckles are metal and quite heavy (Figure 6). Initially, this seems like a negative trait, but things can change further in the design process, as previously mentioned.



Figure 6

The current recommendation would be to use Strap System 1 over system 2, with the possible exception of the chest, where the main PCB would be located.

PCB PLACEMENT

Testing began on the forearm, then the upper arm, then the legs, and finally, the chest. The goal was to place them in good locations for data collection while still providing comfort for the user.

There are two potential locations for the forearm, and having them both could be recommended to help with the tracking of elbow position as well. The goal was to find a flat portion of skin that remains largely

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flat to place the PCB on to prevent migration of the PCB to keep results accurate. A flat surface also provides more comfort for the user, as a regularly deforming portion of skin could get uncomfortable under a PCB. The first is on the back of the forearm just above the wrist, and the second is on the back of the forearm just below the elbow (Figure 7).

The outer portion of the lower arm is significantly less likely to collide with the user's torso than the inside during use. Both regions remain flat for different reasons. The back of the wrist is a largely static portion of the forearm, rotating with the hand when the hand rotates, undergoing little deformation. The outer portion of the forearm remains flat while the radius and ulna twist underneath it when twisting the hand, while the muscle slightly flexes, minimizing rotational motion of the PCB about the lower arm.

These positions are also far enough away from the joints and path of motion when flexion occurs to not interfere with their motion. The rotated position of the forearm is depicted in Figure 8. The full **ROM (range of motion)** is in Figure 9.



Figure 7



Figure 8

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Figure 9

The position of the upper arm PBC is also shown in Figure 9. It was chosen for similar reasons to the ones on the lower arm. That portion of the bicep is out of the way of the torso and remains flat, since it does not sit directly on top of the biceps, triceps, or brachialis muscles. The extended arm position was also tested for consistency (Figure 10).

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Figure 10

The next concern was the legs. Following the same design goals of compatibility with full ROM and muscle flexion, one PCB was placed on the outside of the shin (Figure 11) and the other on the top of the thigh (Figure 12). The full ROM was tested with good results regarding potential discomfort (Figure 13).



Figure 11



Figure 12



Figure 13

One extra option exists for the lower leg PCB placement, and it depends on the inclusion of compression socks in the design. They would go in the same location, but would likely be connected to the footwear instead of the bodysuit (Figure 14).



Figure 14

The last PCB tested was the chest location, which has the Main PCB instead of a Node. The goal was to place it where movement of the head would not collide with the PCB, as well as where it would interfere with spinal flexion (bending forward). This placed it near the center of the sternum, where pectoral flexion was least likely to interfere with the PCB as the shoulders are moved (Figure 15).

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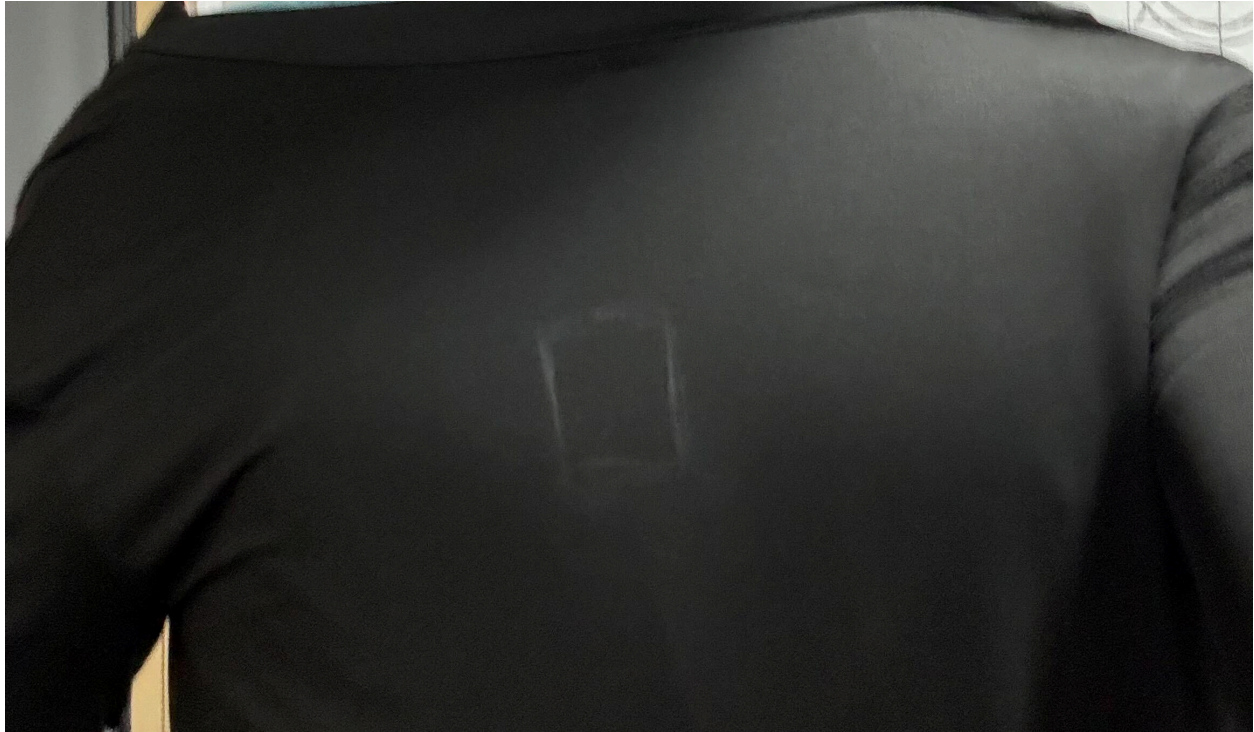


Figure 15

FINAL NOTES

The potential of strap systems was tested, and Strap System 1 showed the most promise. Other untested attachment methods are still under consideration.

The location of PCBs, especially nodes, was tentatively determined. Further research into the requirements of placement to ensure accurate motion tracking is recommended. Placement of nodes near joints or placement of at least two nodes between joints to allow for the determination of the physical location of joints should be considered. Another option is to allow the user to input their personal physical proportion data, allowing for fewer sensors to be required to accurately track user motion. These concerns cannot be definitively solved until a later point in the design and testing processes, but should be kept in mind.

The compression socks, while a viable option, are tentatively not recommended. Most users will have their own footwear, and keeping the design of ExoSkeleton compatible with potential use cases is recommended.